7Aa DOCTORS PAST AND PRESENT

1 Doctors use tests to see if there are changes in a person's body compared to normal. List three tests a doctor may perform.

i _____

ii _____

iii _____

2 A doctor tells a patient that they have acne. Suggest what evidence the doctor has found to make them think this.

_____

3 What does 'evidence' mean? Tick (✓) *one* box.

- ☐ **A** the name given to a certain type of disease
- ☐ **B** changes in the way that the body works
- ☐ **C** information used to say whether something is right or wrong
- ☐ **D** a type of test performed by a doctor

4a Which of the following best describes the heart? Tick (✓) *one* box.

- ☐ **A** an organ
- ☐ **B** a cell
- ☐ **C** a tissue
- ☐ **D** a system

b State one job that the heart does. _____

5 In groups, discuss how to tell if something is alive. Record your group's ideas in the first box below. Ignore the other box for the moment! There is no right or wrong answer.

Ideas from my group:

_____

_____

_____

What we now think:

_____

_____

_____

7Aa LIFE PROCESSES

1 Tick (✓) the living organisms in this list.

- ☐ elephant
- ☐ soccer ball
- ☐ robot
- ☐ pine tree
- ☐ aeroplane
- ☐ human
- ☐ desert
- ☐ mouse

2 A mnemonic (pronounced '*nem-**on**-ic*') is a word or phrase that helps you to remember a list. MRS GREN is a mnemonic to remember the life processes.

a What is a life process? _____

_____

b Draw *one* line from each letter to a life process. Then draw *one* line from each life process to its meanin One has been done for you.

	Life process	Meaning
M	nutrition	increasing in size
R	respiration	getting rid of waste
S	reproduction	making new life
G	growth	releasing energy
R	movement	changing position
E	sensitivity	detecting changes in the surroundings
N	excretion	needing substances to carry on living

3 What do humans need to respire? Tick (✓) *two* boxes.

- ☐ oxygen
- ☐ carbon dioxide
- ☐ wood
- ☐ fire
- ☐ food
- ☐ light

4 Describe one way in which the growth of trees is different from the growth of humans.

_____

SB **5a** In what ways is a car like an organism? _____

_____

b Why is a car not an organism? _____

_____

6 In your group, look at your answers to question **5** on page 3. Discuss your ideas again to find out if the have changed. If you can, write a better answer in the lower box.

ORGANS

1 Use a *pencil* to tick (✓) *one* box to answer to each question.

a Which organ lets you think?
- ☐ **A** heart
- ☐ **B** brain
- ☐ **C** skull
- ☐ **D** liver

b Which organ pumps blood?
- ☐ **A** heart
- ☐ **B** kidney
- ☐ **C** lungs
- ☐ **D** liver

c A job of the stomach is to:
- ☐ **A** store waste materials.
- ☐ **B** make urine.
- ☐ **C** break up food.
- ☐ **D** help you breathe.

d A job of the liver is to:
- ☐ **A** destroy some substances.
- ☐ **B** absorb food.
- ☐ **C** make urine.
- ☐ **D** excrete water.

e A sheep's lungs excrete:
- ☐ **A** mucus.
- ☐ **B** carbon dioxide.
- ☐ **C** urine.
- ☐ **D** oxygen.

f The tube taking food to the stomach is the:
- ☐ **A** intestine.
- ☐ **B** diaphragm.
- ☐ **C** oesophagus.
- ☐ **D** trachea.

2 Ask your teacher how many of your answers to question **1** are correct. Then work with others and the Student Book to find your incorrect answers. Make corrections and check with your teacher. Do this until they are all correct.

3 Give the name of the organ that humans use for sensitivity. _____

4 Give the name of the organ that plants use for nutrition. _____

5 State two functions of a plant root.

i _____

ii _____

6a What process produces the food stored in plant storage organs?

_____

b Why won't a potato grow if the potato plant does not get much light?

_____

_____

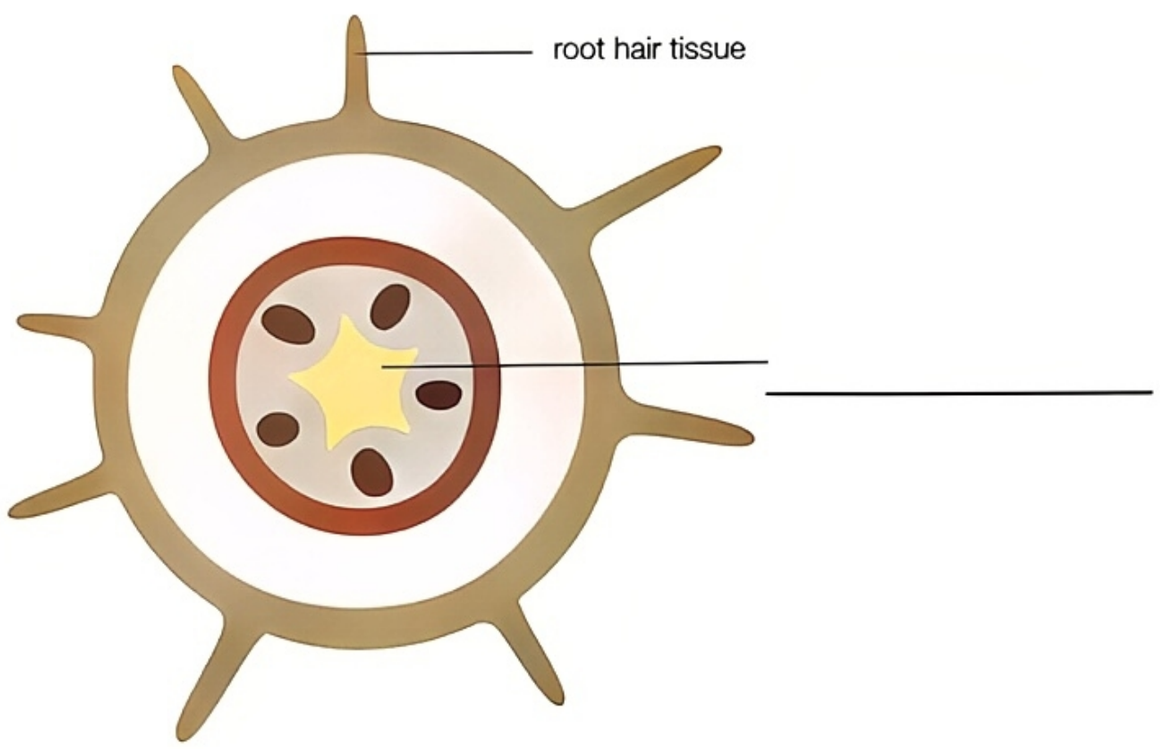
(AC TISSUES

1 Complete the sentences using words from the box. Use each word more than once.

organ tissue

Every _____ in your body is made of many different types of _____. An example of a _____ is fat. An example of an important _____ in your body is your liver.

2 The diagram shows a slice through a plant root.



a What is the function of root hair tissue?

_____

b On the diagram, write in the name of the central tissue (which carries water in a plant).

c How many tissues in total are found in this root?

_____

3 Explain how you can identify one of the tissues in your tongue. To answer this, tick (✓) *one* box for each of parts **a** and **b** below.

a A tissue in your tongue is:

- ☐ **A** hair.
- ☐ **B** root hair tissue.
- ☐ **C** bone.
- ☐ **D** muscle.

b You know this because:

- ☐ **A** your tongue is hard.
- ☐ **B** your tongue feels furry.
- ☐ **C** you can move your tongue.
- ☐ **D** your tongue is wet.

4 The leaf of a plant is an organ.

a Give a function of a leaf. _____

b Suggest one tissue that it contains. _____

c Explain why you think it contains this tissue. _____

_____

5 Complete the table.

Organ	Example of tissue	Function of tissue
heart		

6 What sort of organ is a carrot? _____

7AC MICROSCOPES (WS)

1 Describe what a microscope is used for.

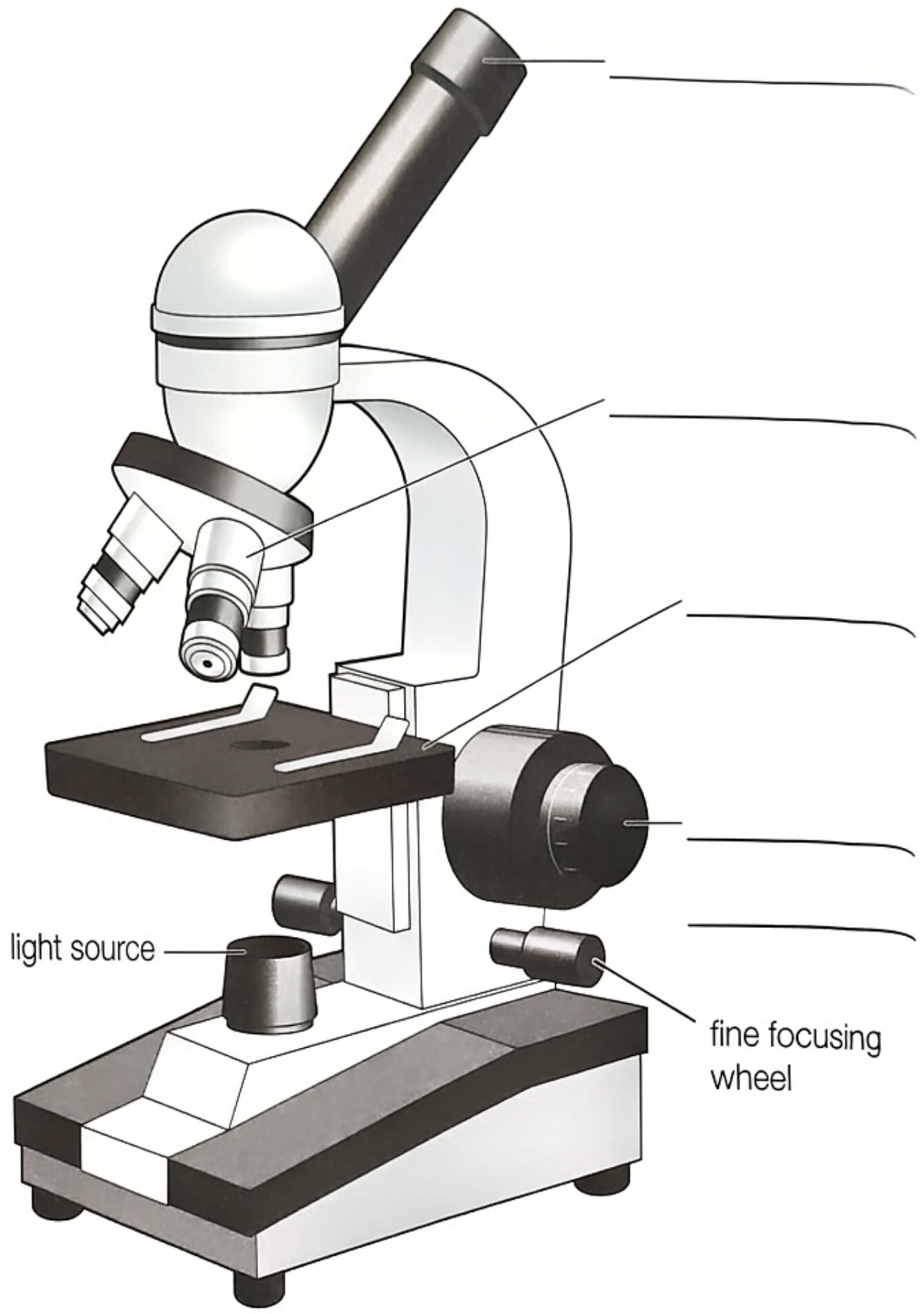
_____

_____

_____

_____

2 Complete the labels on the microscope, to show the names of the parts.



3a Number these sentences in order, to describe how to use a microscope.

☐ Look down a microscope with both eyes open and adjust the light source.

☐ Make the gap between the stage and the lens as small as possible.

☐ Place a slide on the stage. Use the clips to hold the slide in place.

☐ Move the lowest power lens over the hole in the stage.

☐ To magnify more, use the next most powerful lens. Use the fine focusing wheel to adjust the focus

☐ Widen the gap between the stage and the lens, until what you see is clear (in focus).

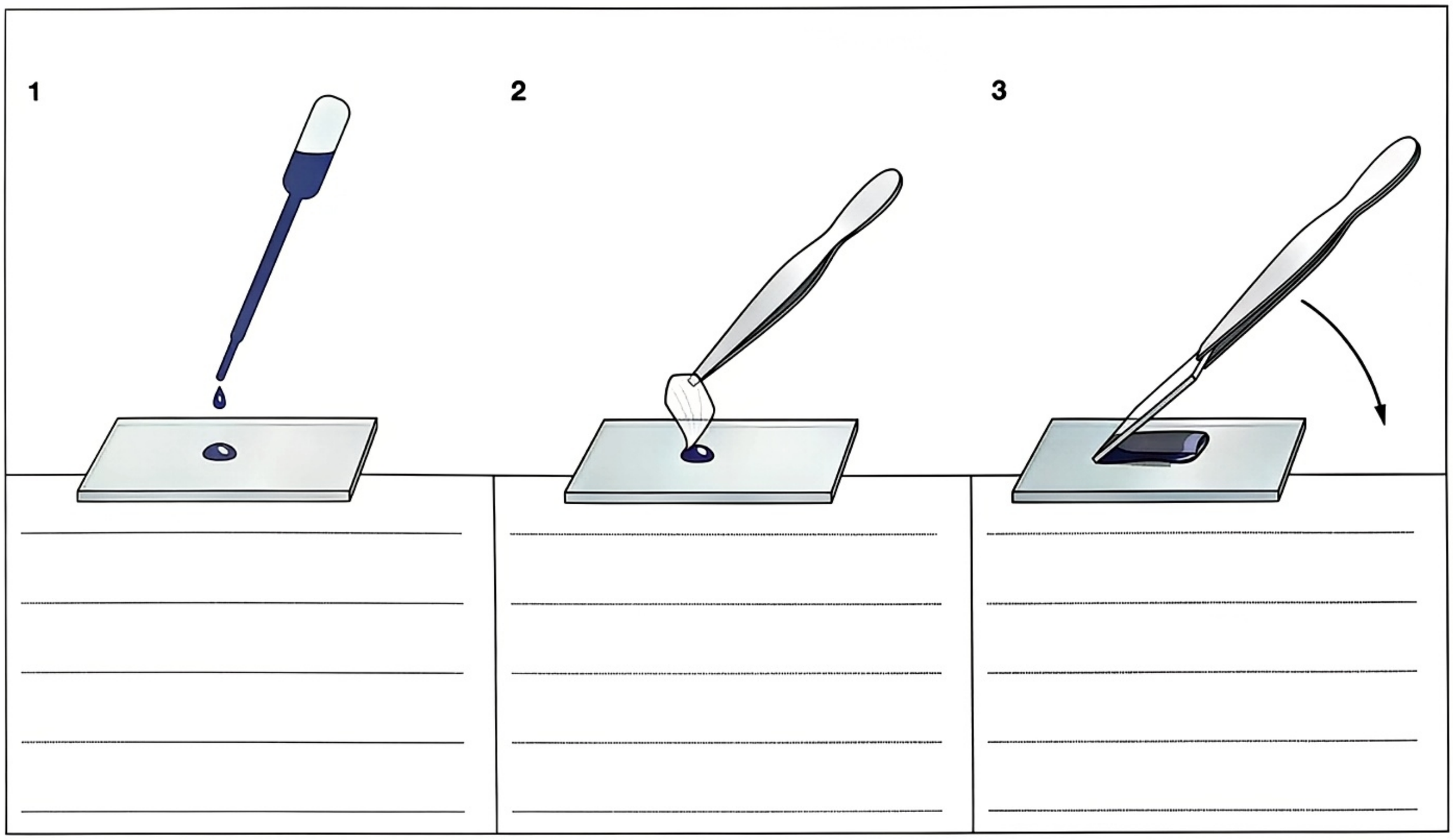
b Describe how you would 'widen the gap', between the stage and the lens.

_____

_____

4 What is a specimen? _____

5a Write instructions for these diagrams to describe how to make a slide.



b Share your instructions with others. Discuss which instructions are best and why. On the lines below, write an improved version of *one* of your instructions.

c How have you improved this instruction? _____

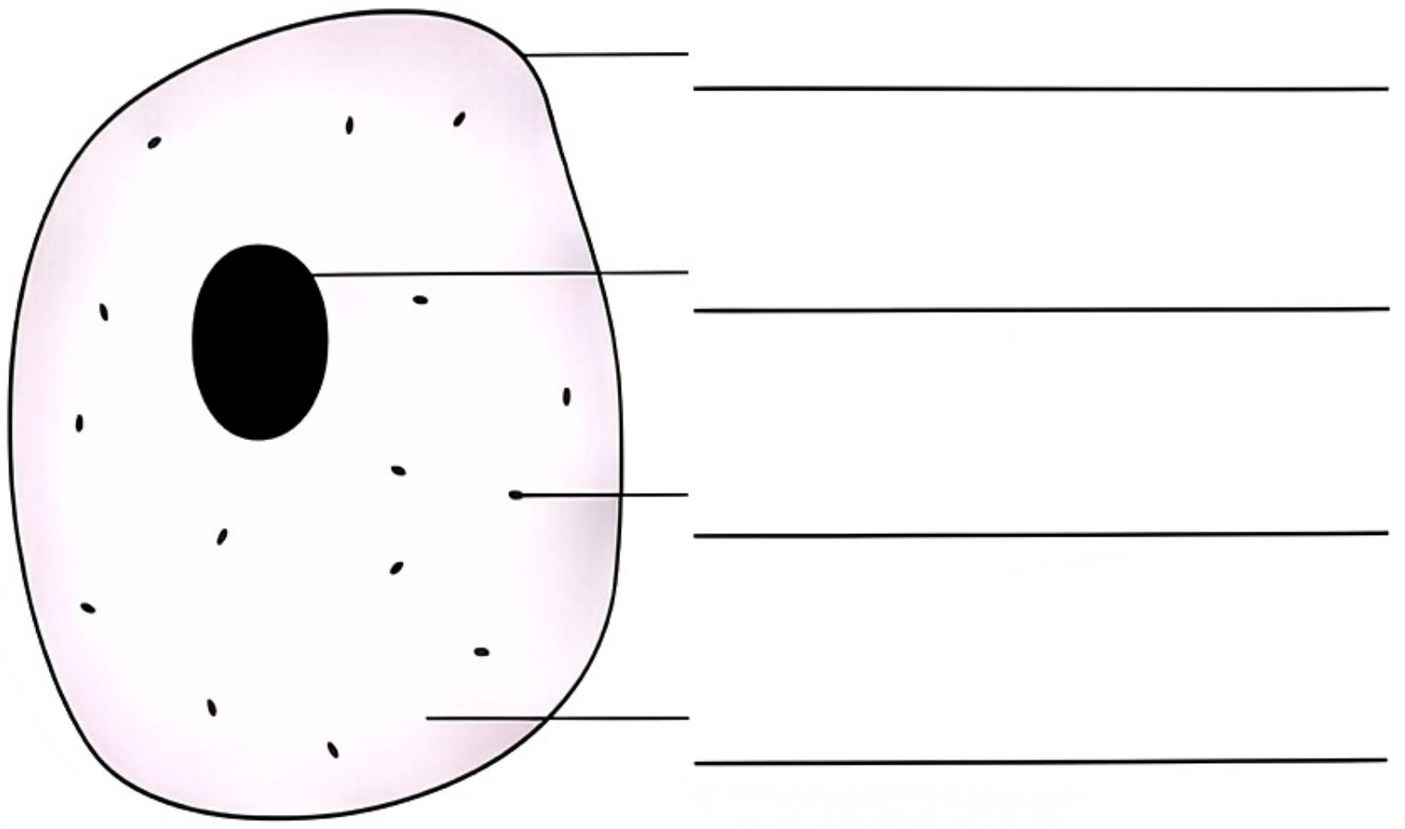
6 Why do we use coverslips? _____

7 Complete the table.

Eyepiece lens power	Objective lens power	Total magnification
×5	×10	
×7.5	×15	
	×20	×200

7Ad CELLS

1 The diagram shows an animal cell. Label the parts.



2 Draw *one* line from each plant cell part to its function.

Cell part	Function
chloroplast	controls the cell
large vacuole	rigid box that helps support the cell
nucleus	storage space
cell wall	where photosynthesis occurs

3 Explain why root cells do not have a part that leaf cells have.
To answer this, tick (✓) *one* box for each of parts **a** and **b** below.

a Root cells do not have:

- ☐ **A** chloroplasts.
- ☐ **B** mitochondria.
- ☐ **C** a vacuole.
- ☐ **D** cell walls.

b This is because:

- ☐ **A** the roots need to be flexible.
- ☐ **B** there is no light underground.
- ☐ **C** root cells do not respire.
- ☐ **D** roots do not store substances.

4 A cell is magnified ×600. What does this mean? Tick (✓) *one* box.

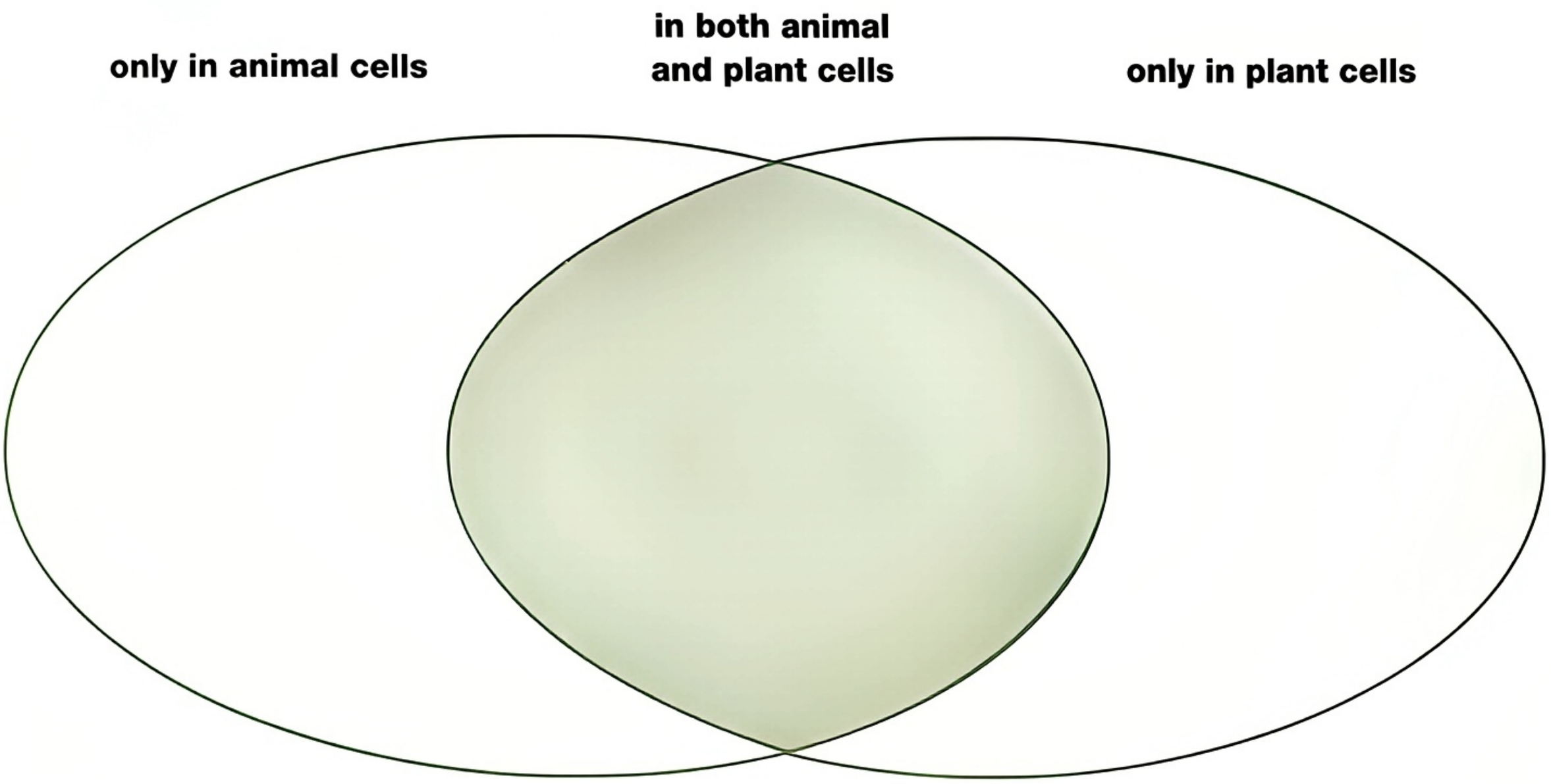
- ☐ **A** It looks 600 times smaller than its real size.
- ☐ **B** There are now 600 cells instead of one cell.
- ☐ **C** It looks 600 times bigger than its real size.
- ☐ **D** Only one cell out of 600 is shown.

SB

5 What makes some plant cells green? _____

6 In what part of a cell would you find cellulose? _____

7 Complete the Venn diagram using the names of cell parts.



8 A leaf cell from a *Eucalyptus* tree is 0.012 mm wide. Calculate its width at a magnification of ×5000.

9a Discuss with others in your group how to draw a labelled diagram of a root hair cell. What parts would you include? How would you do your drawing?

b Draw and label a root hair cell in the space below.